Python List Deep Dive

1 — What is a List?
A list is a **mutable**, **ordered**, and **dynamic** collection in Python.
It can store **heterogeneous** items (ints, strings, objects).
Use it for ordered collections where you will add/remove/modify elements.

List Properties:
- Mutable → Yes
- Dynamic → Yes
- Insertion order preserved → Yes
- Duplicates allowed → Yes
- Heterogeneous → Yes

```
list1 = [1, 2, 3, 4, 5, "123", True, 10+20j]  # Heterogeneous
print(list1)

list1[0] = 100
print(list1)

print(len(list1))
list1.append(1000)
print(list1)

list2 = [1, 2, 3, 1, 2, 3]
print(list2)

# Reading values from list
print(list2[0])  # 1
print(list2[1])  # 2
print(list2[2])  # 3

# Iterating list
for value in list1:
    print(value)
```

Concatenation & Slicing:
Lists can be combined or sliced easily using + and [] operators.

```
list3 = list2 + [10, 20, 30]
print(f"The new list is {list3}")

lst = [1, 2]
print(lst * 5)

integer_list = [1, 2, 3, 4, 1, 2, 3, 4, "abc", "xyz", "aaa"]
print(integer_list[0:4])
print(integer_list.index("abc"))
print(integer_list[8:])
```

Mutable Example:
Lists allow element modification after creation.

```
fruits = ["apple", "banana", "cherry"]
fruits[0] = "orange"
```

```
print(fruits)
fruits.append("Apple")
print(fruits)
```

Validating list & creating sublists:
You can use append() to classify data into different lists based on type.

```
list1 = [1, "abc", 2, 3, 4, 12.35, "xyz", "bbc", 8.36, 2.21]

int_list = []
string_list = []
float_list = []

for value in list1:
    if type(value) == int:
        int_list.append(value)
    elif type(value) == str:
        string_list.append(value)
    else:
        float_list.append(value)

print(f"int_list - {int_list}")
print(f"string_list - {string_list}")
print(f"float_list - {float_list}")
```

Insert, Append, and Extend:
These are the main methods to add elements to lists.

```
list1.insert(2, 1000)
print(list1)

list2 = [10, 20, 30]
list1.append(list2)
print(list1)

list3 = [10, 20, 30]
list1.extend(list3)
print(list1)
```

Removing Elements:
Three common methods: remove(), pop(), and del.

```
fruits = ["apple", "banana", "cherry", "orange"]
fruits.remove("orange")
print(fruits)

fruits.pop(2)
print(fruits)

del fruits[1]
print(fruits)

del fruits[0]
print(fruits)

del fruits
```

Summary:
■ **append()** — Add elements at the end (Best for most cases)
■ **extend()** — Merge lists
■ **insert()** — Add at a specific position
■ **remove()** — Delete by value
■ **pop()** — Delete by index
■ **del** — Delete multiple or entire list